Exam Date & Time: 03-Jan-2025 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

BASICS OF CELL AND MOLECULAR BIOLOGY [BME 3125]

Marks: 50 **Duration: 180 mins.**

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Answer all the questions. Section Duration: 180 mins

Answer all questions

1A)	Examine the roles of the endoplasmic reticulum (ER) and Golgi apparatus in protein synthesis and trafficking. How do these organelles work together to process and transport proteins within the cell?	(3)
1B)	In what ways do carbohydrates support cellular processes in molecular biology? Discuss their roles in energy storage, cell signaling, and structural support, and how their structures enable these functions.	(3)
1C)	What are the key enzymes involved in DNA replication, and how does each enzyme contribute to the process of synthesizing a new DNA strand?	(4)
2A)	What is the role of the mitotic spindle during cell division, and how do its components ensure proper chromosome alignment and segregation during mitosis?	(4)
2B)	Explain the process of nuclear excision repair in DNA. How does this mechanism correct damaged or mutated bases, and why is it essential for maintaining genomic stability?	(3)
2C)	Explain the central dogma of molecular biology and describe the different types of mRNA involved in the process of transcription and translation. How does each type contribute to protein synthesis?	(3)
3A)	Compare and contrast miRNA and siRNA in terms of their structure, function, and mechanisms of action in gene regulation. How do they differ in their roles in RNA silencing and post-transcriptional regulation?	(4)
3B)	Describe the semi-conservative model of DNA replication. How does this model ensure that each new DNA molecule consists of one original strand and one newly synthesized strand?	(3)
3C)	Discuss the mechanisms of epigenetic regulation of gene expression. How do modifications such as DNA methylation and histone acetylation influence gene activity without changing the underlying DNA sequence?	(3)
4A)	Given the pre-mRNA sequence (5' - Exon 1 - Intron 1 - Exon 2 - Intron 2 - Exon 3 - Intron 3 - Exon 4 - 3'), compare the potential protein isoforms produced through alternative splicing. How might different inclusions or exclusions of exons affect the	(3)

function of the resulting protein?

4B)	What are the key steps involved in isolating RNA from a cell? Describe the purpose of each step and explain how different techniques help to ensure purity and yield of the extracted RNA.	(3)
4C)	How does vertical SDS-PAGE work to separate proteins based on their size, and what are the advantages of using this technique in protein analysis? Explain the role of the SDS detergent and the electric field in the separation process.	(4)
5A)	What are the key steps involved in the PCR process, and how does each step (denaturation, annealing, and extension) contribute to the amplification of DNA?	(3)
5B)	Distinguish between a tumour and cancer. Explain the key differences in their growth patterns, potential for metastasis, and their impact on surrounding tissues.	(4)
5C)	Explain what plasmids are and describe the key components of a plasmid. How do these components, such as the origin of replication, selection markers, and cloning sites, contribute to the functionality of plasmids in genetic engineering?	(3)

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